Multirate Variational Methods for Simulation and Optimal Control of Flexible Spacecraft ⋆

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Abstract: We present a multirate variational optimal control method for simultaneous slewing and vibration control in flexible spacecraft. By discretising dynamics on coupled micro and macro time grids, the method preserves conservation properties of the continuous-time model while achieving accuracy comparable to singlerate formulations at significantly lower computational cost. Two multirate formulations are investigated: one that evolves both configuration and momentum, and a configuration-only variant that further reduces the number of unknowns and constraints for improved computational efficiency. Numerical studies of a single-axis manoeuvre demonstrate accurate tracking and effective vibration suppression, with performance matching singlerate methods while substantially reducing problem size and solution time.

Keywords: numerical integration, optimal control, variational calculus, spacecraft control

1. INTRODUCTION

Modern satellites deliver increasingly complex and efficient services under strict requirements on performance, pointing accuracy, and energy consumption. To meet these demands, they often adopt lightweight, flexible designs with appendages such as solar panels, antennas, and robotic manipulators. However, even simple manoeuvres can induce vibrations in these structures, undermining pointing accuracy and potentially affecting structural integrity (Hu et al., 2008; Breakwell, 1981). This highlights the need for computationally efficient control strategies which optimise performance while ensuring safe and reliable operation.

Optimal control has been applied to flexible satellites since the 1970s. Early studies explored performance index selection for a single flexible mode (e.g. Markley, 1979) and feedback control of multiple modes with experimental validation (Breakwell, 1981). Other works compared nonlinear and linearised models (Turner and Junkins, 1980) and considered distributed control systems (Turner and Chun, 1984). In these and many subsequent studies, the satellite is modelled as a distributed parameter system and approximated using spatial discretisation, leading to coupled ordinary and partial differential equations of motion. Two widely used discretisation techniques are the Assumed Modes Method and Finite Element Method (Junkins and Kim, 1993). However, accurately capturing the dynamics of flexible structures with either approach often requires high-dimensional models, posing computational challenges for efficient optimal control implementation.

To address this problem, we propose using Multirate Discrete Mechanics and Optimal Control (Multirate DMOC) (Gail et al., 2017; Junge et al., 2005). In DMOC, the discrete equations of motion are derived from a discrete Lagrange-d'Alembert principle. This yields a structure-preserving scheme that inherits the conservation properties of the continuous-time model and avoids numerical dissipation effects that are common in classical integrators. These equations serve as equality constraints in a discrete time optimal control problem (OCP), leading to structure-preserving necessary optimality conditions. A multirate formulation exploits timescale separation by simulating slow and fast variables on separate time grids while retaining their coupling, thus reducing problem size and computation time relative to singlerate discretisations.

This paper applies Multirate DMOC to the simulation and control of flexible spacecraft, considering a single-axis rotation with quiescent appendages at the end of the manoeuvre. We validate against analytical solutions and compare singlerate and multirate formulations in terms of convergence, conservation properties and computation. We also compare a formulation with explicit momentum and configuration variables to a configuration-only formulation with fewer variables. Spillover due to reduced-order modelling is not considered; our focus is the multirate variational formulation and its balance of accuracy and efficiency.

2. PROBLEM FORMULATION

To represent a wide range of flexible spacecraft, we consider a model consisting of a rigid hub with two identical rigidly attached flexible appendages (Fig. 1). The control input is defined as the torque $\tau(t)$ applied about the hub axis perpendicular to the XY-plane containing the appendages, and we focus on a single-axis rotation manoeuvre. We assume the torque excites the appendages antisymmetrically, producing elastic transverse deformations only in the XY-plane and no shift in the spacecraft's centre of mass. Each appendage is modelled as an Euler-

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Bernoulli cantilever beam with a tip mass, neglecting shear deformation and rotational inertia of the cross section.

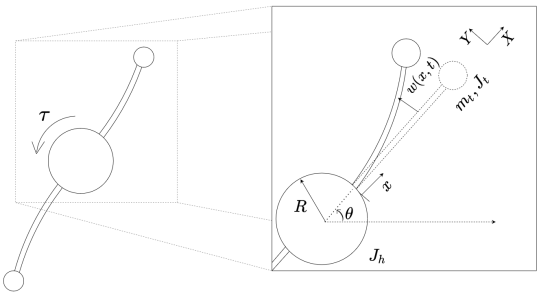


Figure 1. Spacecraft model during single-axis rotation with antisymmetric deformation

A finite-dimensional model is obtained using the Assumed Modes Method (AMM). This expresses transverse deflection of the beam, w, as the sum of N time-varying modal amplitudes, η_j, and assumed spatial mode shapes, ϕ_j,

$$w(x,t) = \sum_{j=1}^{N} \phi_j(x)\,\eta_j(t),$$

$$\phi_j(x) = 1 - \cos\left(\tfrac{j\pi x}{L}\right) + \tfrac{1}{2}(-1)^{(j+1)}\left(\tfrac{j\pi x}{L}\right)^2,$$

where $x \in [0, L]$ denotes position along the beam of length L. Together with the hub angle θ, the amplitudes $(\eta_1(t), \ldots, \eta_N(t))$ form the set of generalised coordinates describing the system configuration. The chosen mode shapes satisfy the boundary conditions of a clamped-free beam and have been shown to produce satisfactory results for tip masses (Junkins and Kim, 1993). Detailed derivations of the linearised equations of motion are provided in Junkins and Kim (1993) and Turner and Chun (1984). The Lagrangian of the resulting system can be expressed

$$\mathcal{L}(\underline{\xi},\, \underline{\dot{\xi}}) = \tfrac{1}{2}\,\underline{\dot{\xi}}^{\intercal}\,\mathbf{M}\,\underline{\dot{\xi}} - \tfrac{1}{2}\,\underline{\xi}^{\intercal}\,\mathbf{K}\,\underline{\xi} \qquad (1)$$

where $\underline{\xi} = \begin{bmatrix}\theta \\ \underline{\eta}\end{bmatrix}$, $\mathbf{M} = \begin{bmatrix} M_{\theta\theta} & M_{\theta\eta}^{\intercal} \\ M_{\theta\eta} & M_{\eta\eta}\end{bmatrix}$, $\mathbf{K} = \begin{bmatrix}0 & 0 \\ 0 & K_{\eta\eta}\end{bmatrix}$, and the time-dependence has been dropped to simplify notation. Here $\underline{\xi}$ represents the configuration vector of the system, $\underline{\eta} = [\eta_1, \ldots, \eta_N]^{\intercal} \in \mathbb{R}^N$, and $\mathbf{M}$, $\mathbf{K}$ are constant matrices.

For this example, the virtual work can be shown to be $\delta W = \underline{\mathfrak{f}} \cdot \delta\underline{\xi} = \tau\delta\theta$ where $\underline{\mathfrak{f}}$ is the vector of generalised forces. Using the Lagrange-d'Alembert Principle, which requires that, for all variations $\delta\underline{\xi}$ with $\delta\underline{\xi}(t_0) = \delta\underline{\xi}(t_f) = 0$,

$$\delta \int_{t_0}^{t_f} \mathcal{L}(\underline{\xi},\, \underline{\dot{\xi}})\, dt + \int_{t_0}^{t_f} (\,\underline{\mathfrak{f}} \cdot \delta\underline{\xi}\,)\, dt = 0, \qquad (2)$$

the spacecraft equations of motion become $\mathbf{M}\underline{\ddot{\xi}} + \mathbf{K}\underline{\xi} = \mathbf{D}\tau$ with $\mathbf{D} = [1, 0, \ldots, 0]^{\intercal}$ (Junkins and Kim, 1993; Ober-Blöbaum et al., 2011). We use the Modal Coordinate Transformation (Meirovitch, 2010; Junkins and Kim, 1993) to decouple the $N+1$ coupled Ordinary Differential Equations (ODEs) forming the equations of motion. This is not required by the multirate method, but is standard practice for problems of this kind. Solving a generalised eigenvalue problem we obtain a set of eigenvalues λ_j and normalised eigenvectors $\underline{e}_j$ satisfying

$$\det(\mathbf{K} - \lambda_j\,\mathbf{M}) = 0, \quad \mathbf{K}\,\underline{e}_j = \lambda_j\,\mathbf{M}\,\underline{e}_j, \quad \lambda_j \le \lambda_{j+1}$$

for $j = 1, 2, \ldots, N+1$ leading to $\mathbf{E}^{\intercal}\mathbf{M}\mathbf{E} = \mathbf{I}$, $\mathbf{E}^{\intercal}\mathbf{K}\mathbf{E} = \mathbf{\Lambda}$, where $\mathbf{E} = [\underline{e}_1, \underline{e}_2, \ldots, \underline{e}_{N+1}]$, $\mathbf{\Lambda} = \mathrm{diag}\{\lambda_1, \ldots, \lambda_{N+1}\}$ and $\mathbf{I}$ is the identity matrix. Physically, each λ_j corresponds to the square of a natural frequency ω_j of the system, and the corresponding eigenvector describes a mode shape. Introducing the linear transformation $\underline{\xi} = \mathbf{E}\underline{q}$ we obtain

$$\mathcal{L} = \tfrac{1}{2}(\,\underline{\dot{q}}^{\intercal}\underline{\dot{q}} - \underline{q}^{\intercal}\mathbf{\Lambda}\,\underline{q}\,), \quad \underline{\ddot{q}} + \mathbf{\Lambda}\,\underline{q} = \mathbf{E}^{\intercal}\mathbf{D}\,\tau, \qquad (3)$$

representing a system of $N+1$ decoupled ODEs, each of which describes a mode with a specific natural frequency.

Having defined the model, we formulate the optimal control problem with the objective of achieving a rest-to-rest single-axis rotation with N quiescent modes at the end of the manoeuvre. The cost function is formulated as a sum of weighted quadratic functions of the modal amplitudes, the modal amplitude rates and the applied control. The optimal control problem (OCP) can therefore be expressed

$$\min_{x,u} \int_{t_0}^{t_f} C(\,\underline{x}(t), u(t))dt = \min_{x,u} \int_{t_0}^{t_f} \underline{x}(t)^{\intercal}\mathbf{W}\underline{x}(t) + u(t)^2 \quad (4)$$

subject to $\underline{\dot{x}}(t) = \mathbf{A}\underline{x}(t) + \mathbf{B}\tau(t)$ and

$$\underline{q}(t_0) = \mathbf{E}^{-1}\underline{\xi}_{\,t_0}, \;\; \underline{\xi}_{\,t_0} = [0, \ldots, 0]^{\intercal}, \;\; \underline{\dot{q}}(t_0) = [0, \ldots, 0]^{\intercal}$$

$$\underline{q}(t_f) = \mathbf{E}^{-1}\underline{\xi}_{\,t_f}, \;\; \underline{\xi}_{\,t_f} = [\theta_{t_f}, 0, \ldots, 0]^{\intercal}, \;\; \underline{\dot{q}}(t_f) = [0, \ldots, 0]^{\intercal}$$

with $u(t) = \tau(t)$, $\underline{x}(t) = \begin{bmatrix}\underline{q}(t) \\ \underline{\dot{q}}(t)\end{bmatrix}$, $\mathbf{A} = \begin{bmatrix}0 & \mathbf{I} \\ -\mathbf{\Lambda} & 0\end{bmatrix}$, $\mathbf{B} = \begin{bmatrix}0 \\ \mathbf{E}^{\intercal}\mathbf{D}\end{bmatrix}$. This formulation of the cost function allows for the minimisation of control effort, while penalising trajectories with large modal deformations, which could lead to potential loss of pointing accuracy, degradation of performance or even structural damage (Hu et al., 2008; Breakwell, 1981).

3. SIMULATION AND OPTIMAL CONTROL USING MULTIRATE DMOC

The system (3) can be separated into slow and fast subsystems based on the natural frequency of each mode:

$$\underline{q} = \begin{bmatrix}\underline{q}^s \\ \underline{q}^f\end{bmatrix}, \qquad \underline{f} = \begin{bmatrix}\underline{f}^s \\ \underline{f}^f\end{bmatrix} = \mathbf{E}^{\intercal}\mathbf{D}\,\tau = \begin{bmatrix}\underline{Z}^s \\ \underline{Z}^f\end{bmatrix}\tau,$$

$$\mathcal{L} = \tfrac{1}{2}(\,(\underline{\dot{q}}^s)^{\intercal}\underline{\dot{q}}^s - (\underline{q}^s)^{\intercal}\mathbf{\Lambda_s}\,\underline{q}^s\,) + \tfrac{1}{2}(\,(\underline{\dot{q}}^f)^{\intercal}\underline{\dot{q}}^f - (\underline{q}^f)^{\intercal}\mathbf{\Lambda_f}\,\underline{q}^f\,),$$

where $\underline{q}^s, \underline{\dot{q}}^s, \underline{f}^s, \underline{Z}^s \in \mathbb{R}^r$, $\underline{q}^f, \underline{\dot{q}}^f, \underline{f}^f, \underline{Z}^f \in \mathbb{R}^{N+1-r}$, $\mathbf{\Lambda_s} = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$, $\mathbf{\Lambda_f} = \mathrm{diag}(\lambda_{r+1}, \ldots, \lambda_{N+1})$, and the time dependence has been dropped to simplify the notation. The size of the slow subsystem r is a free variable and expresses the number of modal coordinates, which will be treated as slow dynamics and thus discretised on a coarser time grid in the multirate approach.

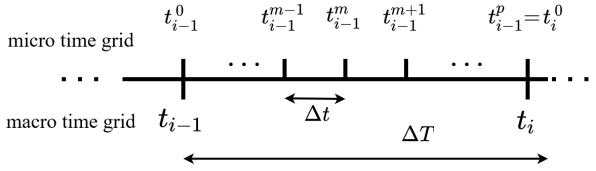


Figure 2. Micro and macro time grid schematic

In the discrete time setting, we introduce two time grids: a macro time grid with the macro time step ΔT and a micro time grid, which is obtained by subdividing each macro step into p equally spaced micro time steps of size Δt as depicted in Figure 2. On the macro time grid, we discretise the slow subsystem, and on the micro time grid, we discretise the fast dynamics. For this purpose, define the discrete paths $\underline{q}_d^s(t_k) = \underline{q}_k^s$ for $t_k = t_0 + k\Delta T$,

$k = 0, \ldots, n_s$ and $\underline{q}_d^f(t_k^m) = \underline{q}_k^{f,m}$ for $t_k^m = t_0 + k\Delta T + m\Delta t$, $k = 0, \ldots, n_s - 1$, $m = 0, \ldots, p$ by approximating the slow and the fast configuration variables using piecewise-linear polynomials where $n_s = t_f/\Delta T$ and denotes the number of macro time steps. On the other hand the control path $u = \tau$ is approximated by piecewise-constant values in $u_d = \tau_d$ which are defined at the midpoints of the micro time grid as $\tau_d = \{\{\tau_k^{m+1/2}\}_{m=0}^{p-1}\}_{k=0}^{n_s-1}$ where $\tau_k^{m+1/2} = \tau(t_k^{m+1/2})$. Additionally, we define $\underline{q}_k^f = \{\underline{q}_k^{f,m}\}_{m=0}^{p}$ and $\tau_k = \{\tau_k^{m+1/2}\}_{m=0}^{p-1}$. The discrete equations of motion are derived following Gail et al. (2017). In direct transcription methods, this is achieved through the direct discretisation of the equations of motion, which we previously obtained using the Lagrange-d'Alembert principle (2). In Multirate DMOC, however, the discretisation is done one step earlier. Namely, the variation principle itself is discretised and the discrete equations of motion are derived from a discrete version of the Lagrange-d'Alembert principle. First, the discrete Lagrangian is defined as an approximation of the action integral for one macro time step

$$\mathcal{L}_k = \mathcal{L}_d(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f) \approx \int_{t_k}^{t_{k+1}} \mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f) dt \quad (5)$$

and thus the discrete action sum is defined as $S_d(\underline{q}_d^s, \underline{q}_d^f) = \sum_{k=0}^{n_s-1} \mathcal{L}_d(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f)$. Similarly, on each macro step, the virtual work of the external forces is approximated as

$$\begin{aligned} \underline{f}_k^{s-} \cdot \delta\underline{q}_k^s + \underline{f}_k^{s+} \cdot \delta\underline{q}_{k+1}^s + \sum_{m=0}^{p-1} (\underline{f}_k^{f,m-} \cdot \delta\underline{q}_k^{f,m} + \underline{f}_k^{m+} \cdot \delta\underline{q}_k^{f,m+1}) \\ = \mathcal{W}_k \approx \int_{t_k}^{t_{k+1}} (\underline{f}^s \cdot \delta\underline{q}^s + \underline{f}^f \cdot \delta\underline{q}^f)\, dt \end{aligned} \quad (6)$$

where the control forces $\underline{f}^f$ and $\underline{f}^s$ have been discretised by discrete left and right forces (Marsden and West, 2001) via $\underline{f}_k^{s\pm} = \underline{Z}^s\left(\frac{\Delta t}{2}\sum_{m=0}^{p-1}\tau_k^{m+1/2}\right)$ for $k = 0, \ldots, n_s - 1$ and $\underline{f}_k^{f,m\pm} = \underline{Z}^f\left(\frac{\Delta t}{2}\,\tau_k^{m+1/2}\right)$ for $k = 0, \ldots, n_s - 1$, and $m = 0, \ldots, p-1$. Thus, for all variations $\delta\underline{q}^s$ and $\delta\underline{q}^f$ vanishing at the end points ($\delta\underline{q}_0^s = \delta\underline{q}_{n_s}^s = \delta\underline{q}_0^{f,0} = \delta\underline{q}_{n_s-1}^{f,p} = 0$) the discrete multirate Lagrange-d'Alembert principle can be expressed as $\delta\sum_{k=0}^{n_s-1}\mathcal{L}_k + \sum_{k=0}^{n_s-1}\mathcal{W}_k = 0$ and the condition for stationary curves $\underline{q}_d^s$, $\underline{q}_d^f$ becomes

$$D_{\underline{q}_k^s}(\mathcal{L}_{k-1} + \mathcal{L}_k) + \underline{f}_k^{s-} + \underline{f}_{k-1}^{s+} = 0 \quad (7a)$$

$$D_{\underline{q}_k^{f,0}}(\mathcal{L}_{k-1} + \mathcal{L}_k) + \underline{f}_k^{f,0-} + \underline{f}_{k-1}^{f,p-1+} = 0 \quad (7b)$$

$$D_{\underline{q}_k^{f,m}}(\mathcal{L}_k) + \underline{f}_k^{f,m-} + \underline{f}_k^{f,m-1+} = 0 \quad (7c)$$

for $k = 1, \ldots, n_s - 1$, $m = 1, \ldots, p-1$. Here $D_{[\cdot]}$ denotes the derivative with respect to the specified argument. These equations are known as the discrete forced multirate Euler-Lagrange equations and describe the dynamics of the system in the discrete multirate setting. Boundary configuration and velocity conditions can be incorporated using the definitions of the fast and slow conjugate momentum, which are obtained using the Legendre transform as shown by Leyendecker and Ober-Blöbaum (2013):

$$\begin{aligned} \underline{p}_k^{s-} &= -D_{\underline{q}_k^s}\mathcal{L}_k - \underline{f}_k^{s-}, \quad \underline{p}_k^{f,0-} = -D_{\underline{q}_k^{f,0}}\mathcal{L}_k - \underline{f}_k^{f,0-} \\ \underline{p}_{k+1}^{s+} &= D_{\underline{q}_{k+1}^s}\mathcal{L}_k + \underline{f}_k^{s+}, \quad \underline{p}_k^{f,p+} = D_{\underline{q}_k^{f,p}}\mathcal{L}_k + \underline{f}_k^{f,p-1+} \end{aligned} \quad (8)$$

for $k = 0, \ldots, n_s - 1$. To obtain a discrete formulation of the OCP (4), Multirate DMOC minimises the cost $\sum_{k=0}^{n_s-1} C_d(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f, \tau_k)$ where

$$C_d(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f, \tau_k) \approx \int_{t_k}^{t_{k+1}} C(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \tau)\, dt \quad (9)$$

is the approximation of the cost functional (4) on each macro time step. The cost is minimised subject to discrete path constraints from (7) and boundary conditions from (8). With slight abuse of terminology, this discrete OCP formulation using Multirate DMOC is termed the Lagrangian formulation (LF), which requires configuration-only integration. Notably, momentum can still be extracted from the LF solution via (8). Replacing (7) with (8) for all k time steps results in the Hamiltonian formulation (HF), which explicitly incorporates momentum simulation. Though theoretically equivalent, LF and HF may converge to different local optima in nonconvex problems, and the choice between them depends on problem objectives and computational resources, as discussed next.

4. COMPUTATIONAL RESULTS

Multirate DMOC was implemented using MATLAB to simulate and optimally control the system with parameters in Table 1 (Junkins and Kim, 1993). The midpoint quadrature rule was used to approximate the relevant integrals in (5), (6), and (9). The model was constructed with $N = 5$ terms in the AMM approximation, and $r = 3$ degrees of freedom were treated as slow dynamics in the multirate formulation. The natural frequencies of the system were obtained as $\omega_1 = 0$, $\omega_2 = 6.454$, $\omega_3 = 52.41$, $\omega_4 = 1.607 \times 10^2$, $\omega_5 = 3.381 \times 10^2$, $\omega_{N+1} = 5.78 \times 10^2$ rad/s, resulting in dynamics on significantly different time-scales and allowing for thorough investigation of the advantages of the proposed multirate scheme.

Table 1. Structural parameters

Parameter	Value	Unit
Hub radius	R	1.0 ft
Hub rotary inertia	J_h	8.0 slug-ft^2
Tip mass	m_t	0.156941 slug
Tip mass rot. inertia	J_t	0.0018 slug-ft^2
Beam length	L	4.0 ft
Beam height	h	6.0 in.
Beam thickness	t	0.125 in.
Beam linear density	ρA	0.0271875 slug/ft
Beam elastic modulus	E	0.1584×10^{10} lb/ft^2

4.1 System simulation

Before implementing the optimal control method, the proposed numerical integration scheme was validated against a reference analytic solution, and its advantages were investigated. Forward simulations were carried out in the absence of control for $t \in [0, t_f]$ starting from a rest position with initial deflection in the appendages of $\underline{\eta}(t_0) = [0.05, 0.001, 0.001, 0.0001, 0.0001]^\intercal$. The numerical convergence of the method was investigated by computing the error of the simulated configuration against the analytical continuous solution at the macro time nodes using

$$e_{\underline{x}}(\underline{x}_d, \underline{x}) = \max_{k\,=\,0,\ldots,n_s} (\|\underline{x}_k - \underline{x}(t_k)\|_\infty) \quad (10)$$

for both the slow and fast variables $(\underline{q}^s, \underline{q}^f)$. Figure 3 presents the results for HF of the approach with $t_f = 1.2\,\mathrm{s}$

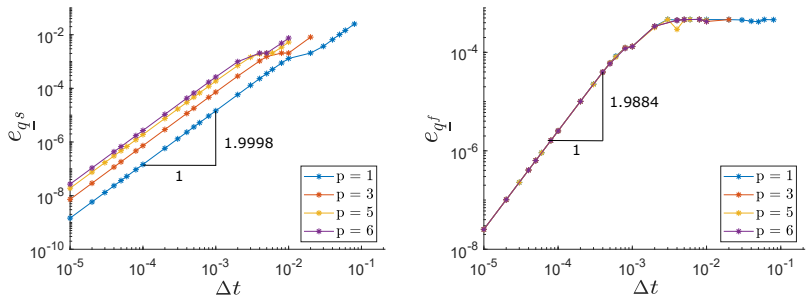


Figure 3. Integrator numerical convergence study using macro-node error computation in HF and $t_f = 1.2\,\mathrm{s}$

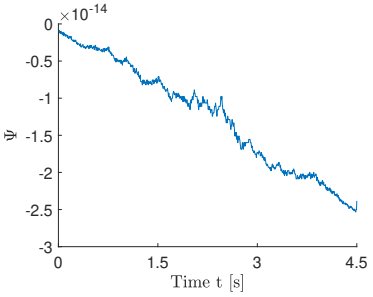


Figure 4. Demonstration of Multirate DMOC conservation properties in HF with $\Delta t = 10^{-3}$, $p = 5$, $t_f = 4.5\,\mathrm{s}$, $\theta_{t_f} = 20°$

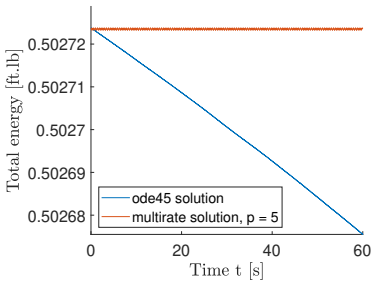


Figure 5. Numerical dissipation of total energy for simulations with $\Delta t = 10^{-4}$ and $t_f = 60\,\mathrm{s}$ in HF

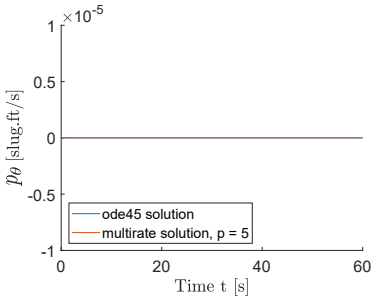


Figure 6. Momentum preservation for simulations with $\Delta t = 10^{-4}$ and $t_f = 60\,\mathrm{s}$ in HF

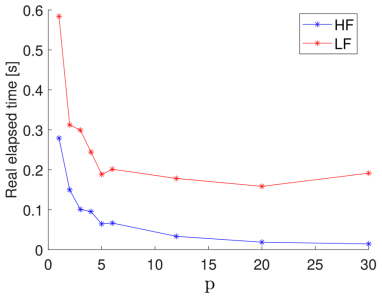


Figure 7. Mean computational time with standard deviation versus p for a constant $\Delta t = 10^{-4}$ and $t_f = 1.2\,\mathrm{s}$

and macro-micro proportionalities of $p = 1, 3, 5, 6$, where $p = 1$ corresponds to a singlerate solution. These figures demonstrate error convergence of approximately order 2, as expected from the order of the chosen quadrature, thus validating the correct behaviour of the numerical integration scheme. Similar results were seen for the LF of the approach. An order reduction is also observed for a small range of time steps, indicated by plateaux, as seen for stiff systems (Simeon, 1998).

Figures 5 and 6 present the evolution of the total energy and the generalised momentum $p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = M_{\theta\theta}\,\dot{\theta} + (M_{\theta\eta})^\intercal\,\dot{\underline{\eta}}$ compared against the *ode45* MATLAB solver ($RelTol = 10^{-10}$) for a simulation with $\Delta t = 10^{-4}$, $p = 5$. These tests use the HF as they required direct momentum computation, though the resulting configuration trajectory closely matches the LF solution, with minor differences arising from implementation specifics, as will be discussed later. Since the Lagrangian (1) is independent of t and θ and no damping or external forces are applied, both the total energy and p_θ should be conserved by Noether's theorem (Marsden and West, 2001). Figure 6 confirms that both *ode45* and the multirate solution preserve p_θ. Figure 5, however, shows that *ode45* introduces numerical dissipation, while the variational scheme preserves total energy up to small bounded fluctuations characteristic of variational integrators (West, 2004). Together, these two tests demonstrate the structure-preserving properties of the variational scheme. Returning to Fig. 3, we can further observe that making the macro grid coarser by increasing p increases the error in the computation of the slow variables slightly, but maintains the same accuracy in fast modes. This small sacrifice in accuracy, however, enables great computational savings. To demonstrate the reduction in computational cost for the multirate approach we performed a series of simulations for a number of macro-to-micro time step proportionalities p while keeping the micro step constant at 10^{-4}. Figure 7 presents the mean of the real elapsed time for the respective simulation obtained from 10 measurements using the *tic-toc* MATLAB routine. Note that we do not focus here on the efficiency of the exact implementation, but rather the comparison between different macro-micro time step formulations. For both HF and LF cases, a clear reduction in computational time is observed as p increases, up to an optimal value ($p = 20$ in this instance for LF). The LF results, in particular, achieve approximately a twofold reduction or more in computational time. The existence of an optimum p-value is due to the specific implementation of the numerical integrator, which uses the Newton-Raphson Method to resolve the system one macro time step at a time. Increasing the number of micro time steps within this period increases the number of variables being obtained at each iteration of the method and thus results in slower computations. Nevertheless, the results demonstrate that significant computational savings are possible for a range of p-values.

4.2 Optimal control using Multirate DMOC

Next, the Multirate DMOC scheme was applied for the solution of the optimal control problem (4) with $\mathbf{W} = \mathbf{I}$. The optimisation was implemented using MATLAB *quadprog* with *interior-point-convex* method and validated against the analytical solution outlined by Turner and Junkins (1980). A numerical convergence test was performed in HF on the macro grid for simulations of length

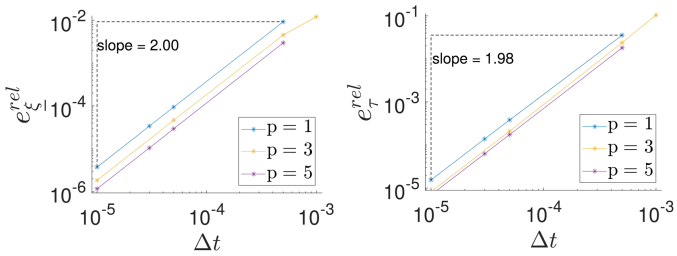


Figure 8. Numerical convergence study (log scale) using macro-node error computation for HF OCP, $t_f = 0.12s$

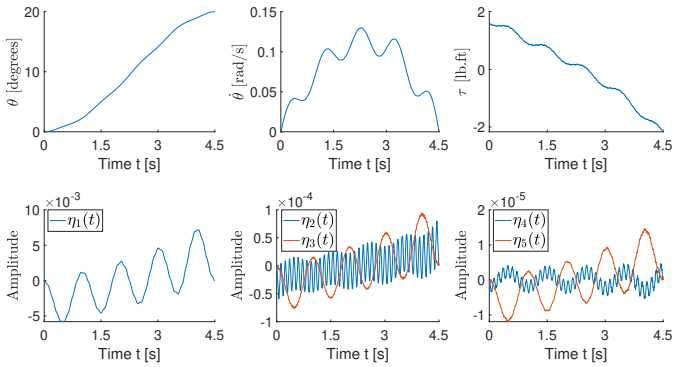


Figure 9. OCP solution with Multirate DMOC for $\theta_{t_f} = 20°$, $t_f = 4.5$ s, $\Delta t = 10^{-3}$, and $p = 5$ in HF

$t_f = 0.12s$ and $\theta_{t_f} = 20°$ using

$$e_{\underline{y}}^{rel}(\underline{y}_d, \underline{y}) = \frac{\max_{k=0,\dots,n_s}(\|\underline{y}_k - \underline{y}(t_k)\|_\infty)}{\max_{k=0,\dots,n_s}(\|\underline{y}(t_k)\|_\infty)} \qquad (11)$$

for the optimal cost C, the trajectory $\underline{\xi}$ and the control force τ. The results presented in Figure 8 are reasonable as performing the manoeuvre in such a short time interval results in values for the cost function and the control input as high as 10^{11} and 10^6, respectively. The figures show convergence of order 2 for all considered variables, as expected from the use of midpoint quadrature, thus validating the correct behaviour of the numerical scheme.

An example solution of the multirate OCP, in its HF, is presented in Figure 9 for a rest-to-rest manoeuvre with $\theta_{t_f} = 20°$, $t_f = 4.5$, $p = 5$, and $\Delta t = 10^{-3}$, demonstrating a successful execution of the manoeuvre and vibration suppression in all modes included in the model. This example solution also demonstrates the structure-preserving properties of the Multirate DMOC scheme. In the presence of control torques, the difference in total energy and external work is now conserved, and the momentum map evolves according to the Discrete Noether's theorem with forcing (Ober-Blöbaum et al., 2011; Marsden and West, 2001). For the current multirate example, this is equivalent to the following discrete conservation law $\Psi_d^k = p_k^\theta - p_0^\theta - \sum_{i=0}^{k}\sum_{m=0}^{p-1} \Delta t \tau_i^{m+1/2} = 0$ for $k = 0, \dots, n_s - 1$ and $\mathbf{E}^\intercal \begin{bmatrix} p_k^\theta \\ \underline{p}_k^\eta \\ \underline{p}_k^\xi \end{bmatrix} = \begin{bmatrix} \underline{p}_k^s \\ \underline{p}_k^f \end{bmatrix}$ based on (3) and $\underline{p}_k^s = \underline{p}_k^{s+} = \underline{p}_k^{s-}$, $\underline{p}_k^f = \underline{p}_k^{f,m+} = \underline{p}_k^{f,m-}$ by substitution of (8) in (7). For the example simulation above, Fig. 4 demonstrates that Multirate DMOC preserves this law, confirming the structure-preserving properties of the multirate OCP formulation.

We illustrate the computational advantages of the multirate OCP scheme in a series of tests comparing singlerate ($p = 1$) and multirate ($p > 1$) solutions. Fundamentally, increasing the proportionality p leads to a coarser macro time grid on which the slow variables are resolved, lowering the accuracy of their computation. On the other hand, using a coarser macro time grid reduces the number of time nodes on which the slow variables are computed, and results in an overall reduction in the number of optimisation variables $n_{total\,var}$ and equality constraints $n_{eq\,con}$ in the optimisation problem. This effect is demonstrated in Fig. 10 for both the HF, used in the OCP tests so far, and the LF, which provides additional savings by eliminating conjugate momentum computation. A series of comparative tests solving the OCP using both formulations is performed, with the quadratic momentum term in the cost zeroed. This is a choice of convenience, as the momentum term in LF can equivalently be expressed via finite-difference velocity approximations without increasing the number of optimisation variables. The figures present simulations with $\Delta t = 10^{-3}$ and $t_f = 4.5s$ and $n_{total\,var} = n_{slow\,var} + n_{fast\,var}$, where $n_{slow\,var}$ and $n_{fast\,var}$ denote the number variables from discretisation on the macro and micro grids, respectively. As shown in Figure 10, compared to the HF, the LF reduces the number of optimisation variables and constraints significantly for all values of p. A more detailed look at the trade-off between accuracy of the simulation and computational cost of the problem is presented in Figure 11. In this test, a series of simulations is performed keeping the micro time step constant at 10^{-3} while varying the macro-micro time step proportionality and recording the resulting variations in both accuracy and real elapsed time for the optimisation. The relative error is computed using Eq. 11 and the computation time denotes the mean time for execution of *quadprog* over 10 instances using the MATLAB *tic-toc* routine. It can be seen that the proposed method can reduce computation significantly with increasing p for small sacrifices in accuracy. For example, in HF the real elapsed time drops by 34% as p increases from 1 to 5 with negligible increase in $e_{\underline{\xi}}^{rel}$. The LF reduces runtime by 61–63% compared to HF across the p range considered, for the same solution accuracy, as expected theoretically. Thus, the multirate scheme allows computation to be traded-off against simulation accuracy.

The trade-off can be optimised by investigating the optimal combination of proportionality p and number of variables r discretised on the macro grid as slow variables. For the HF of Multirate DMOC, the dependence of the number optimisation variables on these choices can be expressed as follows $n_{total\,var} = n_{slow\,var} + n_{fast\,var}$, $n_{slow\,var} = 2r(\frac{t_f}{p\Delta t}+1)$, $n_{fast\,var} = 2(N+1-r)(\frac{t_f}{\Delta t}+1)+\frac{t_f}{\Delta t}$. By increasing r, and thus reducing the number of variables discretised on the micro grid, even larger reductions in computation time are obtained for the same accuracy, as Fig. 12 shows that the relative error curves for different r overlap at the 10^{-5} scale in LF with $\theta_{t_f} = 20°$. Interestingly, $r = 5$ is slightly slower than $r = 3$. As previously discussed, when the slow system increases in size or the macro time step becomes larger, the number of slow variables solved for reduces, but the number of micro steps per macro interval increases, enlarging the fast subsystem solved at each iteration and

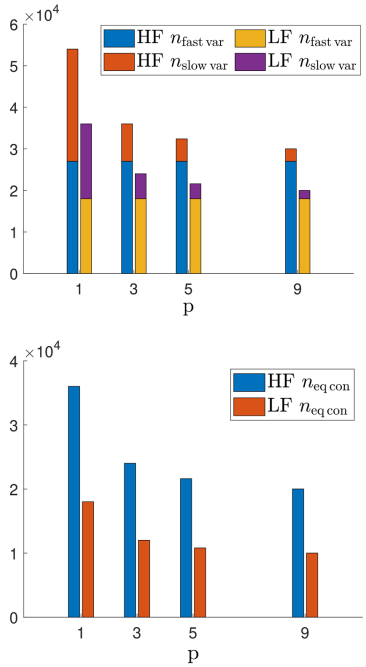


Figure 10. Size of the OCP using Multirate DMOC HF and LF with varying p, $t_f = 4.5$, $\Delta t = 10^{-3}$

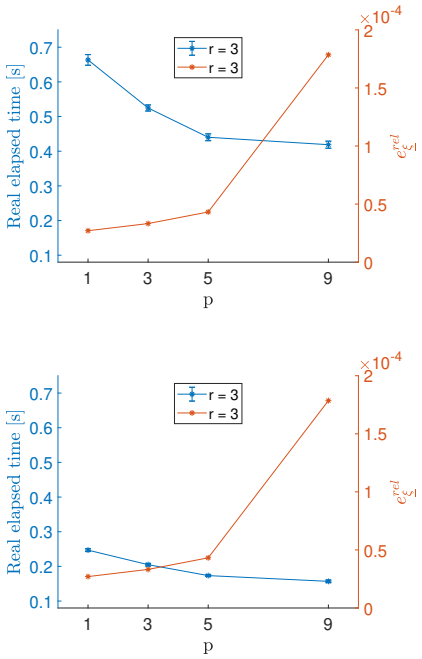


Figure 11. Accuracy (log scale) vs. efficiency using Multirate DMOC HF (top) and LF (bottom) with $t_f = 4.5$, $\Delta t = 10^{-3}$

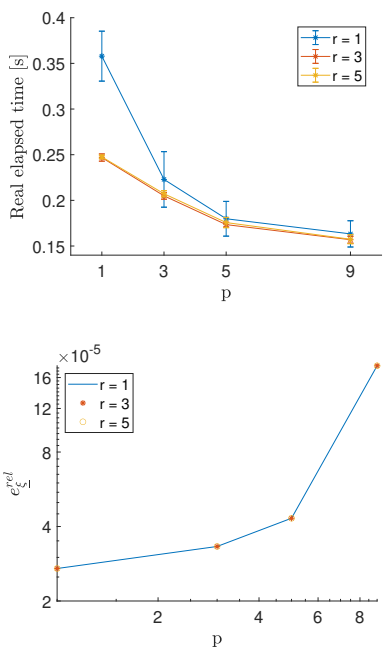


Figure 12. Top: mean runtime vs p with $t_f = 0.45, \Delta t = 10^{-4}$ (top); Bottom: relative error in $\underline{\xi}$ (log scale) vs p, $t_f = 4.5, \Delta t = 10^{-3}$

eventually offsetting the savings. By selecting appropriate p and r, the method can be tailored to the problem's time-scales, providing high-fidelity solutions with reduced runtime.

5. CONCLUSION

We propose a multirate variational optimal control scheme (Multirate DMOC) for simultaneous attitude and vibration control of flexible spacecraft. The method discretises the slow and fast dynamics of the controlled system on macro and micro time grids, respectively. This results in structure-preserving time-stepping equations derived from a discretised variational principle, and removes the need to compute the evolution of the system's velocity. The advantages of multirate DMOC are illustrated using a model of a flexible spacecraft and an optimal control problem consisting of a single-axis rotation with all modes of vibration quiescent at the end of the manoeuvre. Comprehensive investigations for this example demonstrate the numerical convergence and conservation properties of both the numerical integrator and the full multirate OCP scheme.

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